

Gate 4 Reference - technical appendix

Finite strategy comparison and allocation of actual Work

Radiolaria OS / Hedgehog OS · Version 01 · 23 September 2026

Preferences change the selected offer. Experience changes the work allocation. Independent owners retain the decision.

This appendix explains the implemented `G4_REFERENCE_SCOPE_V01` through one controlled Airline story with two causal comparisons. It connects exact mathematical inputs to recorded native Work, consumed outputs and separate current Root reviews. The source anchor is implementation commit `5b259441994ba41ed3467e8ffcb9ae6ee8d371f0`. Presentation extraction reads saved records and source bodies; it does not rerun the project.

The [claim and evidence matrix](#) is the machine-readable companion. It contains exact relative paths, JSON pointers, raw SHA256 bindings, source symbols, test locators, positive and negative routes, consumer routes and slide anchors for claims C01-C16. A test locator names a check in source; it does not mean that the test was freshly executed while preparing this publication.

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A. Result, exact scope and current status

The achieved result has two independent parts. Under the same valid offer set, the declared utility profile changes the selected and prepared offer. Under the same PRICE_FIRST profile and original limits, one admissible history sample changes which seven optional checks actually run. These are operational consequences: the recorded final Work consumes the allocated outputs, and each owner reviews the recommendation separately.

Comparison	Changed factor	Recorded consequence	What remains fixed
CP-STRATEGY	PRICE_FIRST → COMFORT_FIRST utility disclosure	Prepared offer0 → offer1	Two offers, hard constraints, original budget, branch features and limits
CP-BUDGET	Absent prior → one eligible prior of -62,500,000	Optional checks 4/3 → 3/4	PRICE_FIRST, hard constraints, original budget and mandatory work

In CP-BUDGET the prepared offer remains offer0. Adaptation of work allocation is the measured outcome; changing the ticket is not required for this comparison.

Scope statement

Gate 4 Reference implements the owner-approved reduced mathematical scope: finite multi-Root strategy comparison and explainable, bounded allocation of actual work. It reuses the accepted Gate 3 outcome, local-memory and calibration mechanisms; it does not redefine authority or retrain model weights.

The complete Gate 4 programme in Technical Roadmap v2.1 (Phases H-I-J) and Gate-Based Roadmap v3.1 remains planned for reconsideration and continuation after Gate 6 publication, under a separate future scope decision. Deferred work includes unilateral regret and strategic stability, robust/minimax and CVaR, sensitivity analysis, the full AVF feature catalogue, richer alignment priors and broader cross-domain strategic calibration. LGT is deferred and excluded from the completion claim. No delivery date or automatic execution is implied.

G45 records the source installation as

G45_REVIEWED_OWNER_LANDING_COMPLETE_FOR_CLOSURE_REVIEW, with commit

5b259441994ba41ed3467e8ffcb9ae6ee8d371f0, sole parent

50b3df07ef4a1eddb50b3f958592161cea64dee6, and tree

921f07b8ac54e117e2b70e29d3da41ea373bcc62. The delta is 26 additions and 8 modifications. Its normal push, remote readback and installed supplied proof are recorded facts. The Git commit is unsigned.

Independent engineering closure, presentation acceptance and a future documentation publication are separate decisions. The capsule's dated review/status bridge records the current review outcome; archived G40-G44 PENDING, NOT_CLOSED and G45 next-review wording remain unchanged. READY_TO_PUBLISH and REMOTE_PUBLISHED must have separate evidence. A reduced-scope closure is not Full Gate 4 or production readiness.

Sources: [G45 report](#), [G45 result](#), [scope contract](#). Claims C01, C16.

B. Mathematics within the operating architecture

Radiolaria organizes bounded semantic inputs, runtime-owned Work and local final authority. Intent and semantic proposals describe the requested work through existing BSEP and Work contracts. Runtime constructs and admits actual Work. Time and source-bound local memory constrain what may be used now. Root remains the local final authority; an advisory ranking does not become a shared permission token.

Component	Responsibility in this story	Boundary preserved
Bounded semantic input	Supplies configured candidate and utility projections	Proposal/evidence does not grant action permission
G4 strategy	Feasibility, IR, Pareto and finite product ranking	Advisory; current Root review required
G4 pressure/allocation	Assigns the remaining finite local Work budget	Original cap, mandatory work and allowed catalogue retained
Host and Work	Enroll, admit, execute and record actual useful checks	Runtime owns execution and cumulative spending
Gate 3 history / LocalDRS	Records attributable experience and supplies eligible current prior	Historical approval does not transfer current authority
ClientRoot / AirlineRoot / BankRoot	Review their own participation	No SuperRoot, consent transfer or common owner of all budgets

The nine-unit cap belongs to one native task under its enrolled local policy. It is not a budget for three external organizations. The common calculation can use bounded disclosures from the participants without owning their decision rights. These disclosures themselves reveal configured interest information; this is not a zero-disclosure mechanism.

The recorded profile continues finite enrolled Work with `max_children = 0`; it demonstrates actual materialization, continuation and consumption. Arbitrary growth of execution depth is architecture context rather than an outcome of this experiment. Here a quantum is a discrete Work check, not a quantum-computing operation.

Four checks have different jobs: schema validation checks shape; mathematical validation checks exact rules; contextual native validation checks source, task, time, usage and consumption; each Root checks local permission and participation. A schema `PASS` does not cover the other three.

The Root reviews are genuine native, in-process reviews with checked cross-root references. This establishes the recorded local mechanism. It is not a new demonstration of internet federation, legal contracting between external firms or protection against arbitrary hostile code sharing the same process or user identity.

Sources: [runtime consumer](#), [Airline adapter](#), [native checkpoint](#). Claims C01, C03, C08, C09.

C. Evidence chronology and source roles

Epoch	Contribution	How this publication uses it
G41	Pure numerical contracts, independent vectors and 110 tests	Recorded numerical evidence; not fresh G44 passes
G42	First native integration	Historical predecessor, including the later identified self-expected budget/prior defect
G43	Original-basis reconstruction, actual allocation-proof input, currentness and retained-history repair	Source-bound controlled native story; 26 nodes / 78 phases
G44	P1 checker origin closure, P2 observation copies, P3 history subject binding; supplied release registration and retained parent validation	Repaired saved verification and reviewed release successor
G45	Exact owner installation, push/readback and one installed-source supplied proof	Implementation source anchor and installed result
Presentation V01	Explanatory deck, appendix, Reader, claim map and data-only navigation checks	Derived publication; no recollection of native authority

The source roles matter. The portable package's `sources/` records producer bodies for the native story. Its `checker_sources/` and `MANIFEST.json` record the G44 checker closure used for the repaired supplied checks. The capsule's `source_snapshot/installed/` contains the 34 installed G45 postimages. The presentation neither executes archived checker code nor relabels historical producer bytes as the current checker.

The G43 report preserves that the actual G42 consumer accepted coherent 6/1 and unauthorized cold-prior inputs during command 0001; no altered Work was installed. G43 repairs the expected-value boundary by reconstructing it from independent original enrollment, policy, source, usage and validated history. G44 adds the P1-P3 provenance repairs. The historical failed attempts are retained with their original source and result; the current claim rests on the repaired paths and positive neighbors.

The implementation master is specification intent, not runtime evidence. Its A/B/C/D utility table and hand budget vectors are `NUMERICAL_REFERENCE_NOT_RUNTIME`. The native story independently supplies its two offers, utility table, Work and Root records. Earlier CLI proposals also do not override the current implementation: `collect` and `render` are `NOT_IMPLEMENTED` in the pinned entrypoint.

Sources: [G43 report retained in G44](#), [G44 report](#), [G45 report](#), [portable manifest](#). Claims C12-C16.

D. Exact data and declared assumptions

The domain profile is G42_CONTROLLED_AIRLINE_VERTICAL_V01. Native describes execution of the program mechanisms. The offers are controlled records, not a current airline catalogue. There are no new live model calls in G4; PRICE_FIRST and COMFORT_FIRST are explicit configurations rather than captured utterances or model responses.

Field	offer0	offer1
Original offer ID	g40:offer:0	g40:offer:1
Price	EUR 790	EUR 815
Seat characteristics	standard; window	extra_legroom; aisle
Baggage included	true	true
Overnight layover	false	false
Configured budget ceiling	EUR 840	EUR 840
Original route dates	12-21 August 2026	12-21 August 2026
Original snapshot TTL	900 seconds	900 seconds

The source snapshot was created at 2026-09-23T13:40:17+00:00; its original travel dates remain the earlier August dates. This preserves the controlled record exactly. It does not claim current inventory or actual seat reservation.

Human alias	Canonical native identity
Client	root:client_os_001
Airline	root:mock_airline_al
Bank	root:mock_bank_a

The mathematical IDs are separate g4_reference:* projections. /basis/mapping in each lane's allocation record maps them to the original native IDs. Saved utility rows sort by canonical IDs; in the actual arrays Bank precedes Client and Airline. The tables here map by identity, never by row position.

Utility normalizations are prospectively configured. Client price margin is $(840 - \text{amount})/100$; the seat feature is 0 for offer0 and 1 for offer1. PRICE_FIRST gives price weight 1 and seat weight 0. COMFORT_FIRST gives price weight 0.1 and extra-legroom weight 0.9. Airline revenue is $\text{amount}/1000$. Bank margin is $(840 - \text{amount})/1000$. These are configured utility units, not euros, probabilities, measured satisfaction or an empirically established bank profit function. Native penalties and disagreement values in this comparison are zero.

Comparability rests on named fixed factors: original domain records, hard constraints, branch features, quota limits, original policy values and mandatory checks. CP-STRATEGY changes utility disclosure; CP-BUDGET changes eligible history. Task, transaction and BSEP IDs, current native time, Host revision and history descent legitimately differ. Offer-scoped preparation consents are separately created and validated for the actual selected offer. It would be incorrect to claim every byte, including consent and transaction, is identical.

Sources: [comparisons and fixed configuration](#), [original domain](#), [cold mapping and inputs](#). Claims C02-C07.

E. Exact finite strategy mathematics

The frozen numerical profile is G4_REFERENCE_NUMERIC_V01: $Q = 1,000,000,000$, equal exponents $\alpha_i = 1$, and epsilon gain $\varepsilon = 1$ fixed-point integer. The evaluator uses exact integer products and signed round-half-even (RHE). To compute RHE of n/d , it rounds $\text{abs}(n)/d$ to nearest integer, resolves an exact half toward the even integer, then restores the sign. No floating-point ranking is used.

Rule	Implemented expression	Scope
Weighted utility	$U_i = \text{RHE}(\sum_j w_{ij} x_{ij} / Q) - \text{risk}_i - \text{latency}_i - \text{coordination}_i$	Implemented here
Feasible set	$F = \{s: \text{every required hard predicate is known TRUE}\}$	Implemented here
Individual rationality	$IR = \{s \mid F: U_i(s) \geq D_i \text{ for every context participant } i\}$	Implemented here
Strict dominance	a dominates b iff every $U_i(a) \geq U_i(b)$ and at least one is strictly greater	Implemented here
Pareto set	IR candidates without a strict dominator	Implemented here
Integer product	$\text{score}(s) = \prod_i \max(U_{i_fp}(s) - D_{i_fp}, 1)$	Implemented here
Ordering	Descending exact product, then stable candidate ID	Implemented here
Outcome/history update	Existing native proof, calibrated update and current history eligibility	Reused Gate 3
Regret/stability, minimax/CVaR, sensitivity	Explicit NOT_IMPLEMENTED, with deferred reason	Deferred

The implementation requires utility and disagreement values for every participant in the context. A `required_roots` subset does not silently remove another participant from IR or the product. The code first preserves exact term numerators, rounds the weighted sum once, and then subtracts the three penalties. Negative gains fail IR before epsilon is considered. Epsilon handles a zero gain; it cannot offset a loss.

Both native offers are feasible, IR and Pareto in both profiles. There is no invented third dominated native ticket. The utility table below displays U_{fp}/Q ; the product displays the exact native integer divided by Q^3 . Display values never determine the ranking.

Profile	Offer	Client U	Airline U	Bank U	Display product
PRICE_FIRST	offer0	0.500	0.790	0.050	0.01975
PRICE_FIRST	offer1	0.250	0.815	0.025	0.00509375
COMFORT_FIRST	offer0	0.050	0.790	0.050	0.001975
COMFORT_FIRST	offer1	0.925	0.815	0.025	0.018846875

The corresponding canonical integer product strings are:

Profile / offer	Exact product integer
PRICE_FIRST / offer0	197500000000000000000000
PRICE_FIRST / offer1	509375000000000000000000
COMFORT_FIRST / offer0	197500000000000000000000
COMFORT_FIRST / offer1	188468750000000000000000

Equal utility vectors do not dominate each other. Scores are calculated for IR candidates; ranking is over Pareto candidates. A product tie uses stable ID ordering. If every IR candidate has zero gain for every participant, the explicit profile rule returns **NO_DEAL_ALL_ZERO_GAIN**. A recommendation can otherwise have weak IR, so strict improvement for every party is not promised.

Condition	Result
Missing required source, utility, disagreement or suitability	NEEDS_MORE_EVIDENCE; no recommendation
Known predicates leave no feasible candidate	NO_DEAL_NO_FEASIBLE
Feasible candidates all fail IR	NO_DEAL_BELOW_DISAGREEMENT
Every IR gain is zero	NO_DEAL_ALL_ZERO_GAIN
Complete valid nonzero-gain candidate set	RECOMMENDATION with current Root review required
Malformed type, bounds, source or context	Explicit contract failure; not a substantive no-deal

The strict contracts bound participants to 2-8, candidates to 1-32, and utility terms to at most 8 per participant/candidate. Utility features lie in $[-Q, Q]$, weights in $[0, Q]$ and sum to Q ; each penalty lies in $[0, Q]$; utility and disagreement lie in $[-4Q, Q]$. Product strings have at most 80 digits. Canonical inputs are bounded to 262,144 bytes, reports to 1,048,576 bytes; depth and node limits apply before expensive mathematics. Floats, bool-as-int, duplicate keys/IDs, unknown profiles, extra fields and cycles are refused.

The numerical fixture A/B/C/D illustrates dominance separately: $A = (.9, .1, .5)$, $B = (.5, .5, .5)$, $C = (.4, .4, .4)$, $D = (.95, -.1, .6)$, with zero disagreements. D fails IR; B dominates C; Pareto is $\{A, B\}$; products are .045 and .125 and B wins. This is NUMERICAL_REFERENCE_NOT_RUNTIME, covered by the recorded math test `test_strategy_fixed_sets_products_and_witness`.

This is a finite reference bargaining criterion. It is not a Nash equilibrium, strategic-regret calculation, truthfulness mechanism, universal optimum or fairness proof. The fixed scale, epsilon and profile are part of the identity; invariance under arbitrary changes of units is not claimed.

Sources: [strategy source](#), [strict contracts](#), [math tests](#), [cold report](#), [contrast report](#). Claims C04, C05.

F. Pressure, exact apportionment and a real budget

The five-feature profile uses a linear score:

$$z_{fp} = \text{RHE}((4 \times \text{relevance}_{fp} + 2 \times \text{lineage}_{fp} - 2 \times \text{uncertainty}_{fp} - \text{cost}_{fp} + \text{raw_prior}_{fp}) / 8).$$

Relevance, lineage, uncertainty and cost are in $[0, Q]$; the signed raw prior is in $[-Q, Q]$. The exact resulting range is $[-500,000,000, 875,000,000]$. The report retains each signed numerator and the single final rounding. The prior is applied once; an already adjusted G3 score is not added again.

Feature	offer0	offer1	Interpretation
Relevance	.8	.8	Configured bounded inspection relevance
Lineage	.8	.8	Configured source-lineage quality
Uncertainty	.5	.5	Configured lack of certainty; not failure probability
Cost	.4	.43	Estimated normalized per-branch cost; not measured CPU
Cold prior	0	0	No currently eligible sample
History-informed prior	-.0625	0	One matching native G3 prior for offer0

Hard-ineligible or unavailable branches are removed before the weight denominator. Unknown required evidence remains visible. Tampered or unverified prior is a failure; it cannot be silently replaced by a cold zero. Permitted ABSENT, EXPIRED or NONMATCHING history has value zero. Exploration $p = 0$; the mandatory reservation is not an exploration lane.

For surviving branches, the quantized softmax is:

$$w_i = \text{RHE}(W \times \exp((z_i - z_{\max})/\text{tau}_{fp})), \text{ with } W = 10^{12} \text{ and } \text{tau}_{fp} = 250,000,000.$$

Both z and tau are fixed-point integers, so their ratio needs no extra Q . Exponentiation uses a local Decimal context with precision 80 and half-even rounding; it does not mutate ambient process precision. Subsequent redistribution uses integer weights and exact Fraction arithmetic. This is a deterministic finite approximation of exponential weights, not an exact transcendental computation or a probability model.

Quantity	Cold offer0	Cold offer1	History offer0	History offer1
z_{fp}	425,000,000	421,250,000	417,187,500	421,250,000
Integer weight	1,000,000,000,000	985,111,939,603	983,881,318,977	1,000,000,000,000
Optional allocation	4	3	3	4

The signed prior reduces the offer0 score by $62,500,000 / 8 = 7,812,500$. The numerical crossing is visible, but the evidence claim also requires the changed Work set and consumed outputs in section H.

The task budget has exactly nine dispatch units:

Position	Count	Work
Already spent	1	Mandatory initial_constraint
Available optional	7	Admitted offer checks

Position	Count	Work
Still reserved	1	Mandatory final strategy consumer
Original total	9	ONE_NATIVE_WORK_DISPATCH_UNIT

$A = \text{total} - \text{spent} - \text{reserved_mandatory} = 9 - 1 - 1 = 7$. The initial check is already in spent and is not subtracted twice. The final consumer's actual Work ID is `final_strategy`; configuration calls its mandatory role `final_strategy_consumer`. Each branch has lower quota 2 and upper quota 6. One unit means one admitted native Work dispatch, not CPU seconds, tokens, money or FLOPs. Mathematical evaluation and validation are not invented additional Host units.

The allocator starts at lower bounds and sets $T = \min(A, \sum \text{upper})$. It distributes $T - \sum \text{lower}$ proportionally to positive integer weights using exact rational shares. All branches whose tentative share reaches their remaining capacity are saturated together; the remaining resource is redistributed among the others. Final extras are floored; residual units go to the largest fractional remainders, with stable branch-ID ties and caps respected. Thus every allocation respects its bounds, $\sum \text{allocation} = T \leq A$, and $\text{spent} + \text{reserved} + \sum \text{allocation} \leq \text{total}$. Any remainder $A - T$ stays unallocated.

For cold, the three units above the minima have exact offer0 share `3000000000000/1985111939603` and offer1 share `2955335818809/1985111939603`; offer0 wins the final remainder. With history, offer0 has `2951643956931/1983881318977` and offer1 has `3000000000000/1983881318977`; offer1 wins it. No independent clamp loses or invents units.

Negative available budget produces `BUDGET_INSUFFICIENT`; missing pressure evidence produces `NEEDS_MORE_EVIDENCE`; minima exceeding available budget produce `INFEASIBLE_MINIMA`. Refusals do not install partial success plans. Empty surviving sets preserve the remaining optional budget and mandatory obligations.

Sources: [pressure/allocation source](#), [cold allocation](#), [warm allocation](#), [frozen configuration](#). Claims C07, C08.

G. CP-STRATEGY: the preference reaches a prepared offer

The cold lane uses PRICE_FIRST and the contrast lane COMFORT_FIRST. Native final Work produces the exact strategy report, then the adapter binds its candidate identity to the original offer and source snapshot. Separate local reviews receive the chosen candidate, terms, Work result and appropriate offer-scoped preparation consent. Current reviews precede the new hold preparation; earlier reviews remain historical.

Lane	Product winner	Current Client	Current Airline	Current Bank	Prepared result
cold / PRICE_FIRST	offer0	ACCEPT	ACCEPT	ACCEPT	offer0; EUR 790
contrast / COMFORT_FIRST	offer1	ACCEPT	ACCEPT	ACCEPT	offer1; EUR 815
Missing Bank consent control	Recommendation retained	ACCEPT	ACCEPT	NEEDS_USER	No hold continuation

The positive MultiRoot outcome is **PASS**; the missing-consent outcome is **MIXED**. The Bank reason is **user_permission_missing**; the control status is **NEEDS_USER_NO_CONTINUATION**. Permission creation, authority transfer and real-world effect counts remain zero. This distinction shows that mathematical preference and local willingness to participate are separate.

The positive records preserve the complete operational bridge: strategy report → actual Work artifact → current local Root decisions → original offer resolution → offer packet → prepared hold with exact terms and binding. The final status is **PREPARATION_ONLY_NOT_EXECUTED**. No Airline/Bank API call, purchase, debit, booking or seat reservation is asserted.

Sources: [cold consumption](#), [contrast consumption](#), [consent controls](#). Claims C02, C03, C09.

H. CP-BUDGET: experience changes the executed work

The controlled prospective prediction is Boolean FALSE. The real mandatory constraint Work returns a check with `healthy: true`. The prediction is wrong while execution and enforcement are correct. The original output is a TEXT material containing JSON; in `cold/initial_completed.json`, decode `/result/results/0/result/output/0/value` and inspect `/checks/0/healthy`.

Assessment	Saved value	Meaning
Proposal assessment	INCORRECT	The prospective Boolean prediction was wrong
Enforcement outcome	ALLOWED_AS_REQUIRED	The valid required check was allowed
Execution status	COMPLETED	Native Work completed
Task outcome	COMPLETED	Its bounded task completed
Blocking stage	NONE	No Firewall block occurred

The feedback's known `observed_result.value = 0` is the prediction-assessment encoding; it must not be misread as the underlying Work Boolean being false. The saved failure classification `PROPOSAL_INVALID` likewise belongs to this prediction assessment, not an invented malicious action.

Existing G3 capture and proof bind the prospective claim to the actual cold Work original. A separate ordinary recording Root review accepts the exact history snapshot. LocalDRS discovery, eligibility and Root-approved descent make the opened bytes available to the new warm invocation. That snapshot contains raw `prior_fp = -62500000`, `effective_count = 1`, `sample_state = SPARSE`, one negative and zero positive occurrences. The cold allocation cannot consume a sample from its own future.

Optional Work ID	Check meaning	Cold	History-informed
offer_0_1	offer0 route	Executed	Executed
offer_0_2	offer0 baggage	Executed	Executed
offer_0_3	offer0 seat	Executed	Executed
offer_0_4	offer0 price	Executed	-
offer_1_1	offer1 route	Executed	Executed
offer_1_2	offer1 baggage	Executed	Executed
offer_1_3	offer1 seat	Executed	Executed
offer_1_4	offer1 price	-	Executed

The complete finite ordinal catalogue is route, baggage, seat, price, inventory, change_terms for each offer: 12 possible optional quanta, seven performed per task. Six checks are common; the seventh moves between the offers' price checks. The final consumer output enumerates these same checks with their offer IDs, ordinals, field names, input hashes, source digest and checked values. It does not merely repeat an allocation total.

The final capability receives an explicit admitted allocation-proof input. Its result contains `allocation_identity`, `allocation_proof_sha256`, `original_basis_sha256`, `consumed_previous_sha256`, `strategy_identity` and consumed quanta. For cold, the proof digest is `aaa4344b54cfba3342c89f598df1326a9c9cb066312947657a7c2cdb3bafc5c2`; for warm it is

a99afa57b755e9b1b3af4d57517d23b7ca2398e849a9ac7b571e1ee2b3e4140d. These bind actual material and output, with an acyclic dependency rather than a future hash inserted into its own source.

Both cold and warm keep PRICE_FIRST and prepare offer0 after separate current reviews. Each spends exactly nine original units with zero model calls. The three main comparison tasks consume 27 native Work units in total. The independent shorter-offer control spends nine and the historical G40 probe two; those are separate denominators. A terminal retry is UNCHANGED_NO_CHARGE, not another completed check or a replenished cap.

Sources: [prospective/captured G3 material](#), [history recording](#), [warm discovery](#), [cold completed Work](#), [warm completed Work](#), [portable actual Work records](#). Claims C06-C09.

I. Boundary controls with genuine positive neighbors

The most useful controls preserve local consistency and test the relation to an independent original. Outer hashes can be rebuilt while the contextual statement is still false. The refusal should be attributed to the boundary that actually reached it.

Mutation / challenge	Reached boundary and saved result	Positive neighbor
Coherent 6/1 allocation	New consumer constructor: g4_reference:native_source_derived_budget	Normal cold 4/3 installation and completed outputs
USABLE cold prior not yet admitted in history	Same independent native budget reconstruction reason	Valid cold-zero allocation, then valid recorded warm prior
Changed feature or original source identity	Constructor: g4_reference:native_source_derived_budget	Untouched original-basis allocation
Extra quantum, duplicate terminal, foreign definition/material	g4_reference:native_unapproved_work_set	Exact allowed Work set completes
Changed spending revision	g4_reference:native_stale_usage_revision	Current snapshot and original spending accepted
Resealed quota/prior proof	Saved checker: g43_saved:independent_budget_projection	Intact supported saved result
Structurally valid wrong result	g43_saved:root_local_input_binding	Actual final Work in local review
Resealed wrong offer/hold link	g43_saved:selected_offer_hold_link	Original selected offer and bound hold
Claim/native copies disagree with cold originals	g43_saved:observation_cross_copy	Exact original claim and native observation
Genuine same-Root approval for another history subject	P3 subject/meaning binding, tested in test_g44_history_same_root_other_genuine_subject_refuses	Exact snapshot recording review
Missing checker dependency / alternative import origin	P1 installed checker source-closure tests	G44 final and G45 installed supplied checks
Original expiry reached despite fresh review	Actual current consumers: REFUSED; saved event mutation: g43_current_source_expired_or_future	Just-before-expiry prepared hold
Parent raw mutation / coherently resealed semantic FAIL	g44_release:parent_raw_identity:living.json / g44_release:parent_living_public	Untouched parent Living and Conformance PASS

Some source-only test routes deliberately have no invented standalone reason string. The matrix names the actual test locator and retained final focused run. The claim is no broader than that evidence. Native constructor controls preserve the pre-install usage and attempts; their refusal is not drawn as a Firewall block or as a changed executor being allowed to run.

The trusted assumptions are finite and visible: a trusted installed checker, admitted finite capability catalogue, original configuration and source records, the task's clock source, and the local process/Host boundary. Source suitability proves where admitted data came from and how it was used; it does not empirically establish real-world utility truth. The controls make no OS-wide surveillance, arbitrary malicious Python containment or universal security claim.

Sources: [constructor controls](#), [native source controls](#), [final saved-proof controls](#), [parent controls](#), [evidence tests](#). Claims C08-C14.

J. Currentness, trusted checker and supported replay

Current consumption checks the original snapshot, offer and envelope validity using a task-owned real-default or pre-enrollment controlled monotonic clock. A new review or wrapper does not renew the original TTL. The retained tests show acceptance immediately before expiry and refusal at/after it, with future and rollback controls. A separate shorter-offer task provides its own positive neighbor. No machine-clock rewind or artificial lifetime extension is part of the result.

Retained history is first validated at its actual original descent and, when time advances, reopened through a current Root-approved descent. A changed head/prior or expired current opening refuses. The enrolled original proposal, basis, clock and usage remain preserved.

G44's three repairs give the supplied result a distinct trust structure:

Repair	What is independently bound
P1	Actual imported repository and transitive checker origins, finite source ledger, interpreter executable and installed distributions
P2	All copied native/claim/result material to actual cold Work originals
P3	Genuine recording ACCEPT to the exact snapshot, transaction, canonical subject and meaning authority references

The exact recorded checker environment is CPython 3.14.5, built with Clang 21.0.0, jsonschema 4.26.0, referencing 0.37.0. The Python executable is 52,448 bytes with SHA256 2477b47fa3ae65b9574eb18a15edb364e96948eaa1875ad3f1c80d780efc9c12. The verifier checks equality with recorded environment and actual imported origins. Archive files cannot choose replacement code. A generic clean-clone command on any computer is therefore not promised.

Supplied verification is `SAFE_DERIVED_SAVED_RELATIONSHIPS_AT_RECORDED_EVENT_TIMES`. It checks admitted saved relationships with pure validators. Full native canonical replay remains `UNSUPPORTED_NATIVE_SCHEMA`. No Host is deserialized into live authority; no new Root grant is issued; a saved current-at-event-time result does not restore permission today.

The original external test pin is still `TEST_SUPPLIED_PIN`; the presentation must not rename it independent acceptance. A dated independent review can bind the same raw publication bytes separately. Hashes establish byte identity and relationships, not an outside signature, legal certificate or physical truth.

Sources: [checker manifest/environment](#), [G44 checker code](#), [expiry controls](#), [current retained-history controls](#). Claims C10, C11, C13-C15.

K. Verification inventory, measurements and retained limitation

Counts belong to their recorded epochs. Overlapping groups must not be added into a new total, and a presentation-time source read is not a new test run.

Source / run	Scope and lane	Recorded result	Elapsed time
G41 numerical suite	Pure mathematics; historical	110 tests	See original receipt
G43 0009_focused_candidate_02	Native/history/preflight and portable proof	26 nodes; 78 setup/call/teardown phases; 17 native/history/preflight nodes included	60.505123 s
G43 repaired main story	Three controlled native comparison tasks	27 dispatch units	24.392033 s
G44 0017_final_focused	Final focused source/proof/registration	33 PASS / 1 FAIL, 102 phases	45.056284 s
G44 0018_final_parent_controls	Raw and resealed parent negatives with intact positive	Recorded refusals and positive PASS	16.305432 s
G44 0020_supplied_living_final	Pure supplied Living closure	Full canonical result retained	19.888031 s
G44 0021_supplied_conformance_final	Pure supplied Conformance closure	Same full canonical result	19.903165 s
G45 installed supplied	Installed-source pure validation	Same full canonical result	19.859283 s
G44 disposable rehearsal 0023	Installation and ordinary publication A/B states	Recorded finite rehearsal	551.846456 s

The 33 G44 passes contain 14 supplied-proof nodes, 7 registration/parent nodes and 12 selected legacy nodes. They do not include 110 fresh math passes. The G43 26-node set comprises 9 native, 3 history, 9 portable and 5 original preflight nodes; these overlap the cited historical categories.

The one retained FAIL. The unchanged historical mode test requested POSIX 04644, but the environment produced actual 0644. Both old and new verifiers correctly accept actual ordinary 0644. Diagnostic 0019 records the unavailable setuid precondition and equal behavior; it also refuses actual 0666 and symlink substitutions, preserves 0400/0600/0644, distinguishes Git executable identity, and refuses external 0400. The actual policy executable substitution and 21 original restricted files are checked separately in the rehearsal. This remains a test FAIL with an accepted environment limitation, not a fresh PASS, weakened assertion or mathematical defect.

The three final combined outputs are byte-identical at 1,151,634 bytes, SHA256 136b2a76da196e42923886ffc5cf47d89e51d99ec8cc6c7ac549cfaea106b4b9. Whole content equality is stronger than comparing a PASS field. Each named supplied scope reports zero provider/collector/Host reconstruction/Work/history-write/new Root decision/effect/audited network calls. Each G44 combined check performs 430 pure saved Root-result validations including parent checks; those are not 430 new decisions.

Durations are measurements of their exact recorded scopes, not an SLA or CPU billing model. The older G43 replay times near 3.4/3.5 seconds and G44 supplied times near 19.9 seconds cover different work; they do not form a valid speedup/regression benchmark. Nested inclusive durations are not additive CPU shares. The 551.846456-second disposable installation/publication rehearsal is not task latency.

Sources: [G43 report](#), [G44 report](#), [mode diagnostic](#), [G44 Living combined](#), [G44 Conformance combined](#), [G45 installed combined](#). Claim C15.

L. Getting started, detailed verification and deferred work

Begin with the [capsule README](#), then use the [source-bound XML Reader](#) and its detached index. The two-minute reading route is: CP-STRATEGY → CP-BUDGET → actual Work consumer → local Root/consent contrast → source and negative controls. This is a reading route, not a promise about model inference time.

Route	What the reader can establish	Prerequisite
A. Data-only	Completeness, member bytes, Reader/index links, JSON pointers, source/test locators and displayed numerical extraction	Saved capsule and its documented data-only verifier
B. Supplied semantic verification	Supported saved source relationships and recorded-event-time checks	Trusted installed source plus the exact matching recorded checker environment and admitted external trust input
C. Fresh native collection	New native execution under a separately current context	Separate explicit authorization; unnecessary for A or B

The current entrypoint is `demo.run_gate4_reference_v01`. Its documented shapes are `inspect`, `verify --package SAVED_PACKAGE_DIR --trust TRUST_FILE`, and `replay --package SAVED_PACKAGE_DIR --trust TRUST_FILE --output NEW_RESULT`. These are CLI forms, not a claim that this presentation environment ran them. The capsule verification guide distinguishes its actually tested data-only command from the recorded G45 supplied command/receipt. `inspect` is inert but displays historical checkpoint statuses; it is not a current closure query. `collect` and `render` are `NOT_IMPLEMENTED`; `native` and `preflight` really execute and are not needed for presentation extraction.

Do not disable environment/source equality guards to make an arbitrary machine pass. Do not import `checker_sources/` as executable replacement code. If the matching trusted environment is unavailable, route A remains useful and precise while route B retains its recorded result and stated prerequisite.

The original G44 PUBLICATION.json SHA256 is `8e7c4cb6412981436b704d2cc94736de91783d93a77849076007725029d9d08f`; its MANIFEST.json SHA256 is `16cee34833de46c4fd03dee9da3990d03815237567dfec7cc9870f69d6d6f02`. The retained parent manifest SHA256 is `109c37f42ca220782971991c63f40037041d5c4548997fa1ab11262c578c81cd`. The presentation's own detached identities are separate and introduce no self-hash cycle.

Artifact	Role
Main PDF and editable PPTX	One consistent human narrative and active source navigation
This appendix	Exact formulas, units, causal comparisons and methodological limits
XML Reader and detached index	Inert source bodies, evidence and claim navigation
Claim/evidence matrix	Explicit claim → field → source → test → consumer → page routes

Artifact	Role
Evidence and installed source snapshot	Exact original material; producer/checker identities kept distinct
SHA256SUMS, build/QA and status bridge	Packaging identity, actual review, link/visual checks and publication state

Neutral review questions are encouraged: trace the changed prepared offers; identify the exact seven optional Work items in each budget lane; locate the prior producer, recording review and current consumer; test whether Bank consent changes recommendation or use; locate the coherent incorrect allocation refusal; identify any claim whose route is absent or narrower than the wording. XML code and historical prompts are inert analysis material, not instructions to execute or to approve the work.

This publication follows an [artifact-review approach](#), [claim-argument-evidence structure](#), [explicit measurement provenance](#) and [message-led visual evidence](#). Those editorial practices do not confer USENIX, CMU SEI or NIST certification. There is no competitor benchmark, universal safety/fairness claim, quantum execution, fresh live-provider campaign or hidden training result.

The deferred programme remains visible at the end as well as the beginning: unilateral regret and strategic stability; robust/minimax and CVaR; sensitivity; all 17 AVF features; richer alignment priors; broader cross-domain strategic calibration; and LGT. Reconsideration and continuation follow Gate 6 publication only under a separate future scope decision. No delivery date or automatic next-stage execution is implied.

Sources: [pinned CLI](#), [original publication](#), [parent manifest](#), [verification guide](#), [claim matrix](#).